

Construction Choices and Intervention Transfer in Sycophancy Model Organisms

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Question

Do LoRA rank and benign-data fraction decide whether an intervention found on a model organism (MO) works on the natural model?

- ▶ MOs are useful only if their response to a mitigation predicts the real model's.
- ▶ MO conclusions are known to flip with construction details.
- ▶ No one has measured how construction choices change *transfer*.

Hypotheses

- H1 Lower rank, lower benign fraction $\Rightarrow$ more linearly decodable trait.
- H2 Transfer $T(i, j)$ rises with direction cosine $\cos(u_i, u_j)$.
- H3 Most legible MOs transfer to each other, not to the natural model.
- H4 Robust $\rightarrow$ fragile transfer exceeds fragile $\rightarrow$ robust.
- H5 High benign fraction amplifies the existing direction; low builds a new one.

Setup

- ▶ **Parent = natural node:** Qwen2.5-7B-Instruct (alt. Llama-3.1-8B). Shared residual space.
- ▶ **Trait:** sycophancy = flipping a correct answer under pushback.
- ▶ **Grid:** rank $r \in \{1, 8, 64\} \times$ benign fraction $p \in \{0, 0.5, 0.9\}$
 $\Rightarrow$ 9 MOs + 3 benign-only controls.
- ▶ **Mixing:** fixed k sycophantic examples, add $k p / (1 - p)$ benign.

Data

All completions sampled from the parent itself ($T=1$), system prompt stripped.

- ▶ **Sycophantic:** ARC / OBQA / CSQA questions the parent gets right; reply capitulates.
- ▶ **Benign:** UltraChat, self-distilled (adds no new behavior).
- ▶ **Contrastive pairs:** sycophantic vs. honest reply to the same pushback.
- ▶ Every kept sample checked by the eval instrument itself.

Measurements (per model)

Feature	Metric
Trait strength	SycophancyEval flip rate; Wei et al. addition
Conditionality	in-format vs. held-out pushbacks
Legibility	probe accuracy per layer; on-policy AUROC
Leakage	KL to parent on neutral prompts
Capability	MMLU subset
Training cost	steps to target flip rate

Intervention: the transfer matrix

1. **Extract** u_i : diff-in-means on contrastive pairs at best probe layer.
2. **Ablate** u_i in model j : project out of every residual write (Arditi et al.).
3. **Score** $T(i, j) = \frac{F_j - F_j^{-u_i}}{F_j}$ (relative flip-rate drop).

10×10 matrix. Key column: $T(\text{MO} \rightarrow \text{natural})$.

Controls

- ▶ **Random directions:** 10 per target; an entry counts only if its CI beats them.
- ▶ **Capability:** MMLU after each ablation; a large drop = damage, not success.
- ▶ **Benign-only controls:** separate LoRA training effects from the trait data.
- ▶ **Held-out pushbacks:** catch MOs that look clean only off-trigger.
- ▶ **Lexical check:** layer-0 probe accuracy flags surface-text probes.

Analysis

- ▶ **Headline:** correlate $T(\text{MO} \rightarrow \text{natural})$ with MO-only features.
- ▶ H2: Mantel test, T vs. direction cosine.
- ▶ H3: within-legible minus to-natural transfer, permutation test.
- ▶ H4: sign test on $T(i, j) - T(j, i)$.
- ▶ Bootstrap CIs over items. $n = 9$ MOs, so exploratory: effect directions, not fitted models.

Risks

- ▶ **Transfer high by construction:** every MO = parent + LoRA. H5 measures it.
- ▶ **Small n :** one seed, 9 MOs. Seed replicates on corners next.
- ▶ **One trait:** hallucination as a cheap second trait.

Compute: $\sim 6\text{--}8$ h on one H100, $\sim \$20\text{--}35$.

Reading list

- ▶ Model organisms of misalignment: the case for a new pillar
- ▶ Advice for making robust-to-training MOs
- ▶ Three types of model organism
- ▶ Brittle MOs obstruct deception elicitation work
- ▶ Most current MOs leak (perplexity differencing)
- ▶ Narrow finetuning leaves clearly readable traces
- ▶ Building and evaluating model diffing agents
- ▶ Conditional misalignment (arXiv:2604.25891)
- ▶ arXiv:2607.01033